SMART INDIA HACKATHON 2026

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JAALBHEDI



- **Problem Statement ID – SIH26106**
- **Problem Statement Title- AI-Powered Email Threat Detection, GeoLocation and Forensic Intelligence Platform**
- **Theme- Cybersecurity and Blockchain**
- **PS Category- Software**
- **Team ID- 139262**
- **Team Name (Registered on portal) – Sentinel$**

Problem Areas and Proposed Solution

Email Threat & Fraud Detection

Phishing, BEC, impersonation, malicious URLs and attachments.

Sender & Identity Security

Spoofing, lookalike domains and compromised accounts.

Threat Intelligence, Correlation & Attribution

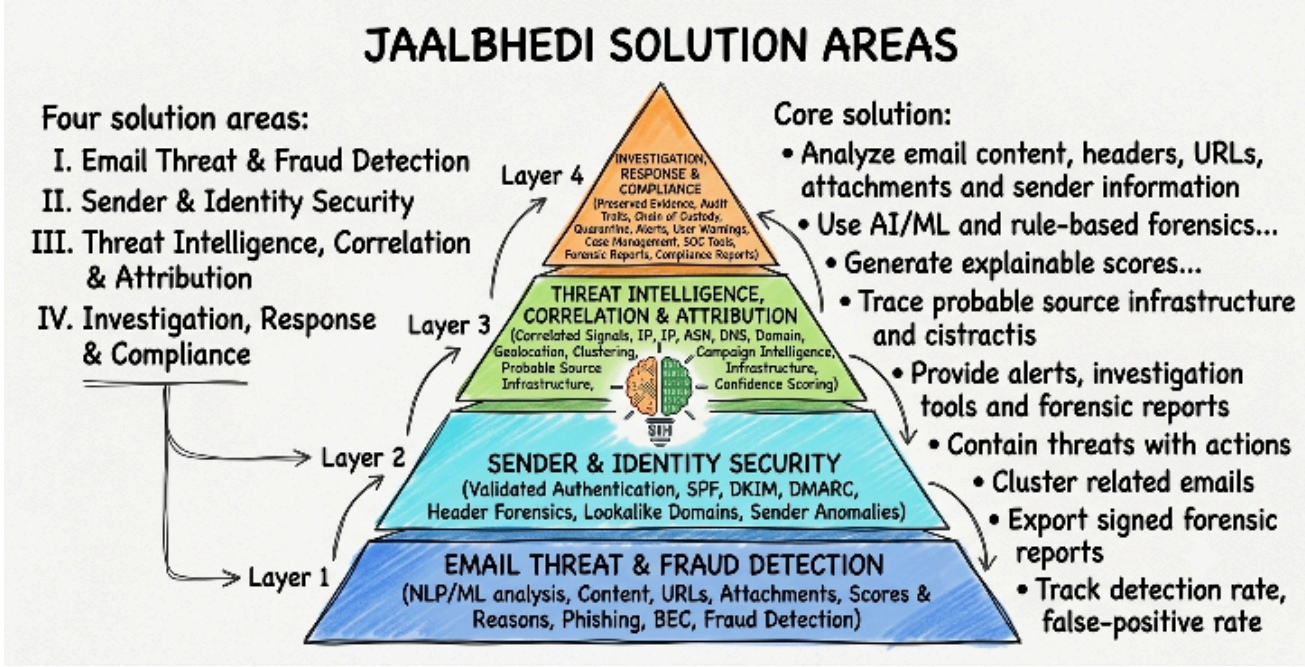
IP, ASN, DNS, domain and geolocation signals can be fragmented or misleading.

Investigation, Response & Compliance

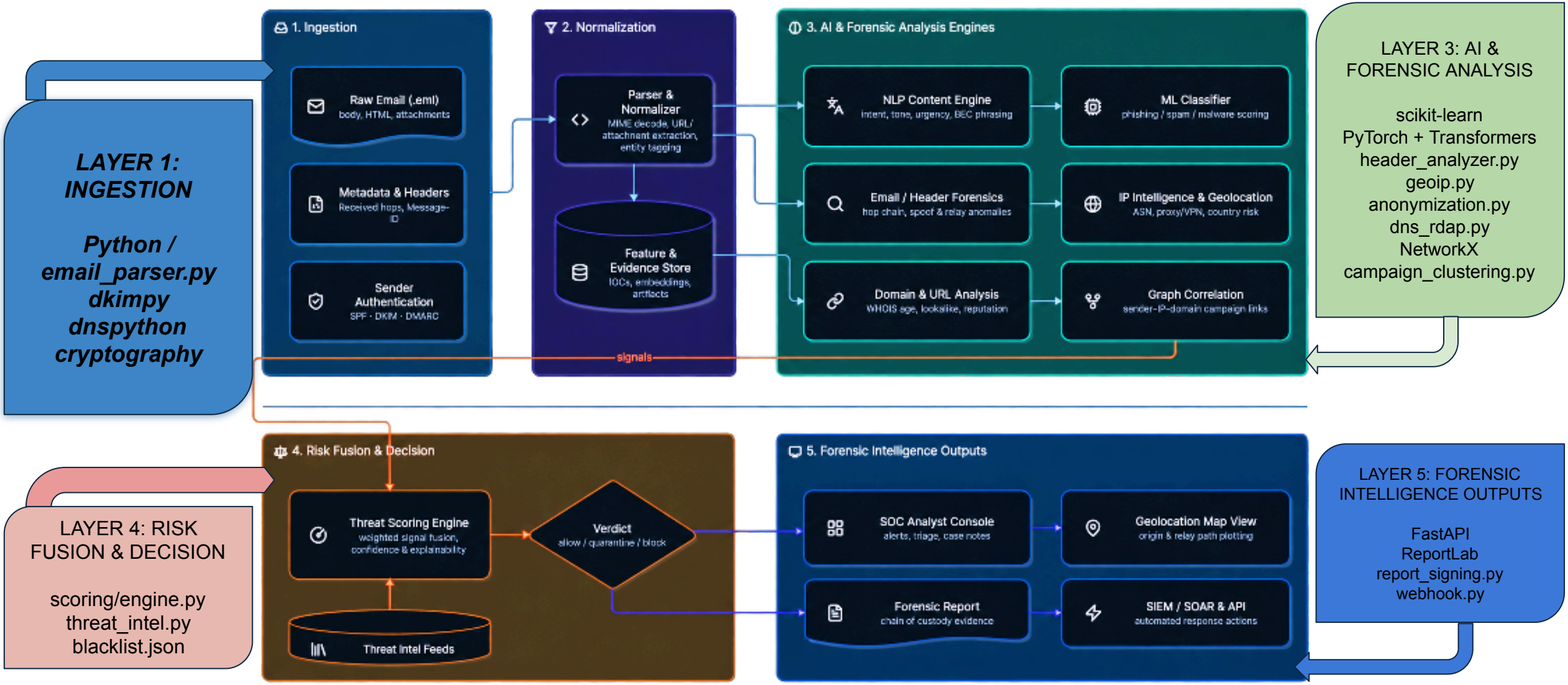
Investigators need preserved evidence, alerts, reports and auditability.

Core solution

- *Analyze email content, headers, URLs, attachments and sender information.*
- *Use AI/ML and rule-based forensics to detect phishing, spoofing, BEC and fraud.*
- *Generate explainable threat and fraud scores.*
- *Trace probable source infrastructure with IP, relay, DNS and geolocation intelligence.*
- *Provide alerts, investigation tools and forensic reports.*
- *Contain threats with quarantine/block actions, user warnings and one-click escalation to an investigation case.*
- *Cluster related emails into campaigns using shared infrastructure, domains and templates to support attribution.*
- *Export signed forensic reports and evidence packages (PDF/JSON) for law enforcement and CERT-In coordination, with case management.*
- *Track detection rate, false-positive rate and time to triage against defined targets, with analyst feedback to improve the model.*



TECHNOLOGY & WORKFLOW



5-Stage Data Flow: Ingestion (raw email, headers, sender auth) -> Normalization (parsing, feature & evidence store) -> AI & Forensic Analysis (NLP/ML, header forensics, IP/geolocation, domain & graph correlation) -> Risk Fusion & Decision (threat scoring, verdict) -> Forensic Intelligence Outputs (SOC console, geolocation map, forensic report, SIEM/SOAR).

Critical Vulnerabilities and Mitigations

#	Critical Vulnerability	Effect on Your System	Solution / Mitigation
1	Incorrect Origin Tracing	Header/IP intelligence may identify a VPN, proxy, cloud server, relay or compromised machine instead of the attacker.	Correlate Received headers, SPF/DKIM/DMARC, IP/ASN, DNS, domain intelligence and VPN/TOR/proxy indicators. Report probable source infrastructure with confidence.
2	Legitimate / Compromised Sender Bypass	A compromised legitimate account may pass SPF, DKIM and DMARC and appear trustworthy to the ML classifier.	Detect behavioral anomalies in sender IP, time, recipient patterns, volume, domains and history. Feed deviations into the Threat Scoring Engine.
3	Malicious Content Attacking the Analyzer	Parser, URL analysis and attachment extraction process attacker-controlled data; crafted inputs can crash or exhaust analysis infrastructure.	Use isolated, sandboxed processing. Enforce file-size, MIME-depth, CPU, memory and timeout limits. Add SSRF protection, redirect limits and network isolation.
4	AI/ML Evasion	AI-generated or modified phishing can make NLP/ML wording look legitimate.	Use layered detection: NLP/ML plus header forensics, authentication, domain/URL and IP intelligence, behavior and graph correlation.
5	Forensic Evidence Tampering	Modified emails, indicators or investigation records can undermine forensic reports and attribution intelligence.	Hash original evidence with SHA-256. Maintain chain of custody, immutable audit logs and timestamps. Protect evidence with encryption and RBAC.
6	Limited Training Data	Scarce labeled phishing, BEC and spoofing samples can limit NLP/ML accuracy and generalization to new attack patterns.	Bootstrap with public phishing/spam corpora and synthetic augmentation. Use active learning so SOC analyst feedback continuously retrains the model.
7	Processing Scale and Latency	High enterprise/government email volume can slow multi-source correlation, delaying triage and alerts.	Use a tiered pipeline: lightweight pre-filter before deep forensic analysis, async queues and horizontally scalable microservices.
8	SOC and SIEM Integration Gaps	SOCs already run SIEM/SOAR and mail gateways; poor interoperability slows adoption of a new platform.	Expose standard REST, Syslog and STIX/TAXII connectors and SIEM/SOAR plugins. Validate through a phased pilot before full rollout.
9	Regulatory and Privacy Compliance	Processing email content and forensic evidence across jurisdictions raises data-protection and evidentiary concerns.	Apply configurable data residency, encryption at rest and in transit, RBAC and audit trails aligned to IT Act 2000 and sectoral rules; legal review for admissibility.
10	Cost and Resource Constraints	Multi-source AI/ML correlation across IP, ASN, DNS and geolocation can be compute-intensive for smaller organizations.	Offer tiered on-prem/SaaS deployment and optimize models (quantization, cached threat-intel lookups) for affordable smaller-SOC editions.

Mitigations preserve attribution confidence, analysis resilience, evidence integrity and overall system feasibility and viability.

Impact, Value Proposition and Beneficiaries

It is only a prototype, not a fully working model.

Impact

- Earlier identification of phishing, spoofing, BEC and fraud.
- Faster triage through explainable scores and correlated indicators.
- Stronger evidence integrity for investigations and reporting.

Value Proposition

- One workflow joins content, header, identity, infrastructure and behavioral analysis.
- Probable source infrastructure with confidence, rather than unsupported exact attribution.
- Security controls support SOC work, auditing and compliance.

Beneficiaries

- Security Operations Centres and incident-response teams.
- Government, enterprises, banks and public-sector organizations.
- Forensic investigators, compliance teams and end users.



Threat verdict, explainable score and evidence fingerprint



Sender authentication, origin path, phishing cues and signed forensic report export

JAALBHEDI focuses on defensible findings and operationally useful evidence.

Research Grounding and Market Comparison

Research and standards references

Research References
• NIST SP 800-177 Rev. 1 – Trustworthy Email
• NIST SP 800-86 – Integrating Forensic Techniques into Incident Response
• RFC 7489 – DMARC
• RFC 6376 – DKIM
• RFC 7208 – SPF
• MITRE ATT&CK T1566 – Phishing
• MITRE ATT&CK T1566.001 – Spearphishing Attachment
• MITRE ATT&CK T1566.002 – Spearphishing Link
• MITRE ATT&CK DET0070 – Phishing Detection Strategy

Literature Survey

- Earlier studies mainly focus on detecting phishing, BEC and spoofing using ML, NLP and email-header analysis.
- Other research looks at email authentication, campaign detection, threat intelligence, explainable AI and digital forensics.
- These approaches provide useful solutions, but are often handled separately.
- Our approach: JAALBHEDI brings these techniques together into one workflow for email detection and forensic investigation.

Prediction

When we simultaneously examine the content of the email, headers and sender characteristics and related infrastructure, we can more effectively detect phishing, spoofing and BEC threats and easily put forth relevant explanations and evidence in order to initiate remedial actions.

Market comparison

Capability	Typical email security tools	JAALBHEDI
I. Email Threat & Fraud Detection	Phishing and malware detection	NLP/ML analysis of content, headers, URLs, attachments and sender data; explainable threat score and reasons
II. Sender & Identity Security	SPF, DKIM, DMARC checks	Validates SPF, DKIM and DMARC; analyzes headers and spoofing; finds lookalike domains; flags sender anomalies
III. Threat Intelligence, Correlation & Attribution	Reputation and blocking signals	Correlates IP, ASN, DNS, domains, geolocation and VPN/TOR/proxy/cloud indicators; clusters related emails into campaigns; probable source infrastructure with confidence
IV. Investigation, Response & Compliance	Alert and investigation context	SHA-256 evidence hashing, chain of custody, quarantine/block and warnings, SOC case management, signed export for law enforcement and CERT-In, RBAC, encryption and audit logs

Differentiation: joins detection, infrastructure correlation and evidence-preserving investigation in one workflow.

References are standards and publicly documented techniques, not claims of feature parity with every commercial product.